Swati Aggrawal

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PROFESSIONAL SUMMARY

MSc Computer Science graduate from Sapienza University of Rome with a strong background in machine learning research, statistical analysis, predictive modeling, and transformer-based AI. Experienced in designing reproducible experiments, benchmarking machine learning models, and analyzing large datasets using Python and PyTorch. Published IEEE researcher with a passion for quantitative research, mathematical modeling, and solving complex analytical problems.

SKILLS

- **Programming:** Python, SQL, Java, C++, Bash
- **Mathematics & Analytics:** Statistics, Linear Algebra, Probability, Data Analysis, Statistical Analysis, Predictive Modeling, Feature Engineering, Model Evaluation
- **Machine Learning:** PyTorch, TensorFlow, Scikit-Learn, Deep Learning, Transformer Models, Vision Transformers, Continual Learning
- **Tools:** Linux, Git, Docker, AWS

EDUCATION

Sapienza Università di Roma **May 2026**

Master of Science in Computer Science

- **Relevant Coursework:** Natural Language Processing, Deep Learning and Applied AI, Machine Learning, Computer Vision, Distributed Systems
- **Thesis:** Evaluating Pretrained Vision Transformers Under Continual Learning.
- Selected for a competitive international master's program.

Sharda University **July 2023**

Bachelor of Technology - CSE with AI and ML

- **Relevant Coursework:** Operating Systems, Data Structures, Machine Learning, Databases, Cloud Computing
- **Thesis:** Virtual Generative Model Inducing Human Identity to Computers in Metaverse
- Awarded **merit-based scholarship** (Top 5–10% of cohort).

PUBLICATION

Machine Learning for Air Pollution – IEEE Conference Publication (2023)

- Evaluated Random Forest, SVM, Decision Tree, and Neural Network models on environmental datasets.
- Performed comparative analysis using accuracy, robustness, and generalization metrics.
- Authored and presented findings through an IEEE peer-reviewed publication.

RESEARCH EXPERIENCE

Evaluation of Pretrained Vision Transformer Architectures Under Continual Learning Settings (Master's Thesis) **Nov 2025 - May 26**

- Conducted research on continual learning for Vision Transformer architectures, evaluating model robustness, catastrophic forgetting, and knowledge retention across sequential learning tasks.
- Reviewed and analyzed recent research literature on continual learning and transformer-based computer vision models to guide experimental design.
- Designed experimental pipelines for training, evaluation, and comparison of computer vision models across sequential tasks.
- Evaluated model performance under changing data distributions and analyzed model reliability.
- Investigated catastrophic forgetting and knowledge retention in transformer-based visual learning systems.
- Developed reproducible training and evaluation pipelines using Python and PyTorch.
- Automated experiment tracking and performance benchmarking across multiple continual learning scenarios.
- Documented experimental findings and identified research opportunities for future development of continual learning systems.
- **Technologies:** Python, PyTorch, Vision Transformers, Computer Vision

PROJECTS

DocDelta AI-Powered Document Intelligence & Comparison Platform

- Developed an AI-powered document intelligence application with backend APIs for automated document analysis and semantic comparison.
- Integrated Sentence Transformer models into inference pipelines for semantic similarity and document intelligence tasks.
- Designed RESTful endpoints using FastAPI for automated document analysis and change tracking workflows.
- Built document ingestion, semantic analysis, and comparison pipelines using transformer-based NLP models.
- Leveraged transformer-based embeddings for semantic document understanding and retrieval-oriented workflows.
- Implemented semantic matching to identify added, removed, and modified content across document versions.
- Containerized the application using Docker and deployed it on AWS ECS Fargate.
- Configured application monitoring and logging using AWS CloudWatch.
- **Technologies:** Python, FastAPI, Sentence Transformers, Docker, AWS ECS Fargate, CloudWatch, REST APIs

Metal Defect Detection using Computer Vision

- Developed computer vision models for automated defect detection using CNN, SVM, and Random Forest.
- Designed and evaluated machine learning pipelines, conducting comparative performance analysis across multiple models.
- Conducted comparative statistical evaluation of deep learning and classical machine learning models.
- Performed model evaluation using accuracy, confusion matrices, and performance analysis.
- **Technologies:** Python, TensorFlow, Keras, OpenCV, Scikit-Learn

Lossy Text Compressor [Semantic NLP Pipeline]

- Investigated semantic text compression using transformer-based embeddings, quantization techniques, and text reconstruction methods.
- Evaluated reconstruction quality using cosine similarity and ROUGE metrics.
- Analyzed trade-offs between compression efficiency and semantic preservation across multiple quantization levels.
- **Technologies:** Python, PyTorch, Transformer Models, NLP, Sentence Embeddings